## SHA-3 Capacity Lanes

**29. Write a C program for SHA-3 option with a block size of 1024 bits and assume that each of the lane in the first message block (P0) has at least one nonzero bit. To start, all of the lanes in the internal state matrix that correspond to the capacity portion of the initial state are all zeros. Show how long it will take before all of these lanes have at least one nonzero bit. Note: Ignore the permutation. That is,**

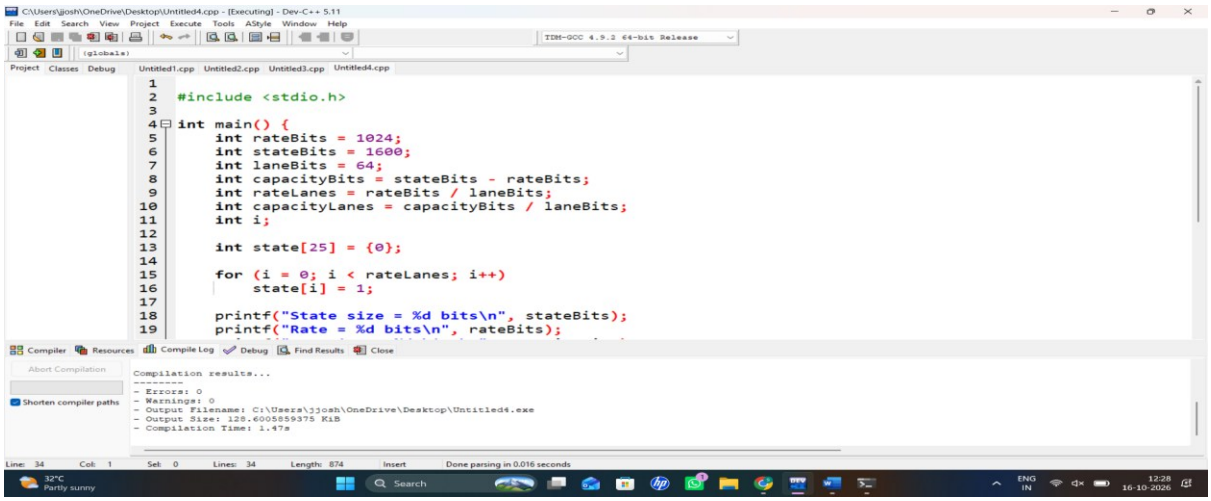
## Aim:

To write a C program to analyze the zero-valued capacity lanes in SHA-3 with a block size of 1024 bits, ignoring the permutation.

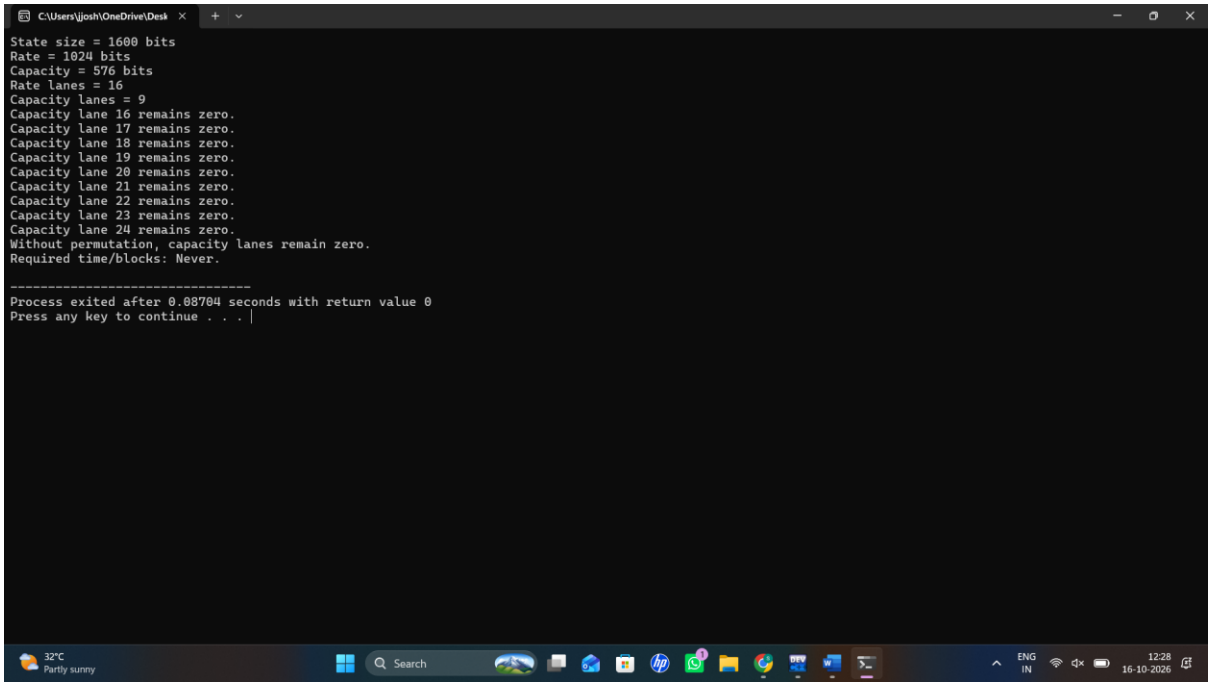
## Algorithm:

1. Consider the SHA-3 internal state size of 1600 bits.
2. Set the rate to 1024 bits and the capacity to \(1600-1024=576\) bits.
3. Divide the state into 64-bit lanes.
4. Initialize the 16 rate lanes as nonzero and the 9 capacity lanes as zero.
5. Ignore the permutation and track the capacity lanes.
6. Determine whether all capacity lanes become nonzero.

## Program:



## Output:




## Result:

The program shows that the 9 capacity lanes remain zero when the permutation is ignored. Therefore, they never become nonzero through absorption alone. In actual SHA-3, the permutation mixes state bits, so this conclusion does not describe the full SHA-3 operation.